

2024



Bitlayer Builds the First Layer 2 BitVM Platform for Seamless Bitcoin Transactions on AWS

Learn how Bitlayer created the world’s first BitVM-based Layer 2 solution for Bitcoin, processing over 150,000 transactions daily and supporting over 300 projects using Amazon EC2.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Outcome](#) | [AWS Services Used](#)

150,000+
average daily transactions processed

300+
projects launched in first six months

members served

reduction in computing costs



Hi, I can connect you with an AWS representative or answer questions you have on AWS.



Need more info? Highlight any text to get an explanation generated with AWS generative AI.



Overview

[Bitlayer](#) focuses on advancing Bitcoin's role in a fully decentralized economy through its innovative BitVM technology. The company sought to develop a scalable, secure solution that could support both on-chain and off-chain transactions while optimizing infrastructure costs. By building its platform on Amazon Web Services (AWS), Bitlayer successfully scaled its operations and enhanced performance to meet the growing demands for smart contracts. The platform now holds over \$550 million in total value locked, supports around 300 live Bitcoin projects, and serves a community of more than 700,000 members.



Opportunity | Becoming the First Layer 2 Bitcoin BitVM Provider

Singapore-based Bitlayer aims to position Bitcoin as the foundation of a fully decentralized economy. Founded in 2023 and valued at over \$300 million, Bitlayer has secured major investments from U.S. financial firms. Kevin He and Charlie Hu co-founded Bitlayer in October 2023 after introducing the Bitcoin Virtual Machine (BitVM). BitVM enhances Bitcoin's computing power while preserving its core principles of security and decentralization. He and Hu recognized Bitcoin's potential to play a role beyond payments—including digital assets, financial services, and online gaming.

Bitlayer set out to become the [first Layer 2 solution on the Bitcoin blockchain](#) to utilize BitVM for both on-chain and off-chain transactions. To achieve this, the company required a scalable and secure infrastructure to support its business model. "Our challenges were two-fold," explains He. "We needed a cloud platform that offered the scalability and security Bitcoin demands, while also helping us manage costs since we're still in the start-up phase."

Solution | Saving 20% on High Performance Computing Costs

Drawing on its prior experience with AWS from earlier Bitcoin projects, Bitlayer chose to build the infrastructure for its BitVM offering on AWS. “We knew AWS was widely used in the Web3 blockchain community for its scalability and global reach,” explains He. “We were also aware of AWS’s efforts to ensure secure infrastructure, which is crucial for building trusted blockchain-based solutions.”

To carry the project through from development to going live, the Bitlayer team set up both its Testnet and Mainnet blockchains on AWS. Initially, the company relied on on-demand [Amazon Elastic Compute Cloud](#) (Amazon EC2) instances for resource-heavy workloads, but it later transitioned to [Amazon EC2 Reserved Instances](#) (RI) for predictable computing needs. “Switching to Amazon EC2 RI helped us cut our compute costs by about 20 percent,” says He. “The AWS team recommended the change, highlighting their focus on cost optimization from the beginning.”

Given the Mainnet environment's need for rapid access to critical data and applications, Bitlayer deployed [Amazon Elastic Block Store](#) for persistent, secure, high-performance block storage. It also used [Amazon Relational Database Service \(Amazon RDS\) for PostgreSQL](#) for its database needs, utilizing [AWS Multi-Availability Zone](#) architecture to safeguard performance. “This setup means we can perform upgrades without slowing down or stopping servers, which is essential for maintaining high availability,” says He.

Outcome | Reaching Over \$550 Million in Platform Assets in Months

Within six months of Bitlayer’s Mainnet launch, the value of Bitcoin assets on the platform had surpassed \$550 million, establishing Bitlayer as a leading Bitcoin Layer 2 solution provider. Daily transactions averaged over 150,000, and monthly website visitors grew to 859,000. “Our choice to build on AWS proved to be the right one,” He affirms. “The performance of our Mainnet scaled effortlessly as the number of transactions reached hundreds of thousands.”

Today, Bitlayer supports a community of more than 700,000 members who have deployed over 300 projects onto the platform. It also attracted \$11 million in series A funding from major US global investment management companies. With this investment, the co-founders are expanding the development team and accelerating the creation of Bitlayer’s Mainnet-V2 to enhance existing capabilities.

With the development of BitVM and the potential for Turing-complete functionalities—the ability to execute any programmable logic, like smart contracts—within the Bitcoin network, Bitlayer aims to expand cryptocurrency’s role



beyond being just digital gold. The company envisions Bitcoin facilitating lending, borrowing, and the automatic execution of smart contracts without intermediaries. “With AWS, Bitlayer is well positioned to drive Bitcoin Layer 2 innovation and help the Bitcoin network securely transfer and store value, unlocking new opportunities for everyone,” He concludes.

About Bitlayer

Founded in 2023 and headquartered in Singapore, Bitlayer is a Bitcoin Layer 2 solution built on BitVM technology. It offers Bitcoin-level security and enhanced computing capabilities, aiming to provide scalable, secure solutions for the Bitcoin ecosystem.

AWS Services Used

Amazon Elastic Compute Cloud

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 750 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

[Learn more »](#)

Amazon EC2 Reserved Instances

Amazon EC2 Reserved Instances (RI) provide a significant discount (up to 72%) compared to On-Demand pricing and provide a capacity reservation when used in a specific Availability Zone.

[Learn more »](#)

Amazon Elastic Block Store

Amazon Elastic Block Store (Amazon EBS) is an easy-to-use, scalable, high-performance block-storage service designed for Amazon Elastic Compute Cloud (Amazon EC2).

[Learn more »](#)

Amazon Relational Database Service (Amazon RDS) for PostgreSQL

Amazon RDS makes it easier to set up, operate, and scale PostgreSQL deployments on the cloud.

[Learn more »](#)



More Customer Stories

HONG KONG



Bitlayer Builds the First Layer 2 BitVM Platform...

Learn how Bitlayer created the world's first BitVM-based Layer 2 solution for

2024

HONG KONG



The Bank of East Asia Improves Its Mobile App...

Learn how The Bank of East Asia, Limited (BEA) delivers a better mobile

2024

HONG KONG



Crypto.com Builds AI Engine to Accelerate...

Crypto.com uses a fully managed infrastructure for generative AI on AWS to build a scalable visual content

2024

HONG KONG



HKBGE Enhances Upcoming Crypto Exchange...

Hong Kong BGE Limited (HKBGE) is developing a generative AI customer service chatbot on Amazon EC2 G5

2024

Get Started

Organizations of all sizes across all industries are transforming their businesses and delivering every day using AWS. Contact our experts and start your own AWS journey today.



Learn About AWS

- What Is AWS?
- What Is Cloud Computing?
- AWS Accessibility
- AWS Inclusion, Diversity & Equity
- What Is DevOps?
- What Is a Container?
- What Is a Data Lake?
- What is Artificial Intelligence (AI)?
- What is Generative AI?
- What is Machine Learning (ML)?
- AWS Cloud Security
- What's New
- Blogs
- Press Releases

Resources for AWS

- Getting Started
- Training and Certification
- AWS Solutions Library
- Architecture Center
- Product and Technical FAQs
- Analyst Reports
- AWS Partners

Developers on AWS

- Developer Center
- SDKs & Tools
- .NET on AWS
- Python on AWS
- Java on AWS
- PHP on AWS
- JavaScript on AWS

Help

- Contact Us
- Get Expert Help
- File a Support Ticket
- AWS re:Post
- Knowledge Center
- AWS Support Overview
- Legal
- AWS Careers



Amazon is an Equal Opportunity Employer: *Minority / Women / Disability / Veteran / Gender Identity / Sexual Orientation / Age.*

Language

- عربي |
- Bahasa Indonesia |
- Deutsch |
- English |
- Español |
- Français |
- Italiano |
- Português |



Tiếng Việt |
Türkçe |
Русский |
ไทย |
日本語 |
한국어 |
中文 (简体) |
中文 (繁體)

Privacy
|
Accessibility
|
Site Terms
|
Cookie Preferences
|

© 2024, Amazon Web Services, Inc. or its affiliates. All rights reserved.

